Non-Destructive Cross-Sectioning Technique For Checking Components

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Acoustic micro-imaging tools are usually pictured scanning flat subjects such as plastic-encapsulated ICs, JEDEC trays filled with components, multilayer ceramic chip capacitors or other flat items. The tools use a horizontally-scanning ultrasonic transducer that pulses ultrasound into the subject thousands of times per second and analyses echoes returned by material interfaces to produce pixels for the acoustic image of the subject.

The tools excel at performing scans to produce top-view images of internal structural features, especially sub-surface and internal gaps (cracks, delaminations, voids, etc) that can cause failures. Planar raster scanning of the sample's x-y area is known as C-mode imaging. The method is non-destructive and a handy compliment to X-ray because it easily provides top-view detection of very thin gaps (down to a small fraction of a micron) that X-ray cannot see.

The acoustic tool can also be used to collect z-dimension data from a vertical plane to produce a non-destructive cross-section through a sample. One advantage of non-destructive cross-sectioning is that the same sample can be sectioned acoustically in as many vertical planes as desired with no damage to the sample. The planes do not need to be parallel to each other; the sample can be rotated as desired and then scanned.

This imaging mode, known as Q-BAM (Quantitative B-scan Analysis Mode), was developed by Sonoscan (now Nordson SONOSCAN) and is used by the company's C-SAM line of acoustic micro-imaging tools. The operator of the tool first defines an acoustic cut line on the top surface of the sample in a location that will give the best information in a cross-sectional view.

Background: C-mode imaging

In horizontal imaging, the transducer scans across the sample in a straight line, advances the width of one line and scans back across the sample. As it scans, the transducer sends thousands of pulses of ultrasound each second into the sample's surface and receives from each pulse the returned echoes, if any, before launching the next pulse. When it has scanned the entire area of the sample, it will have collected all data points needed for the acoustic image.

Within the sample, the pulse of ultrasound is reflected only by material interfaces. Homogeneous materials, such as underfill or silicon, may absorb varying amounts of the ultrasound, but being homogeneous these send back no reflection. Interfaces between two solid materials—mould compound to die face, for example, or die face to underfill, or a crack in a die—send back reflections whose amplitude ranges from very low to very high, depending on the properties of the two materi-